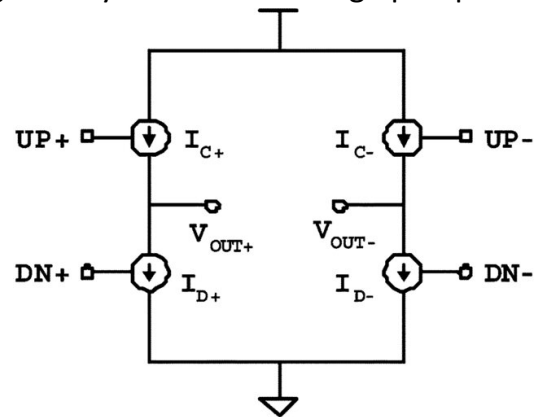


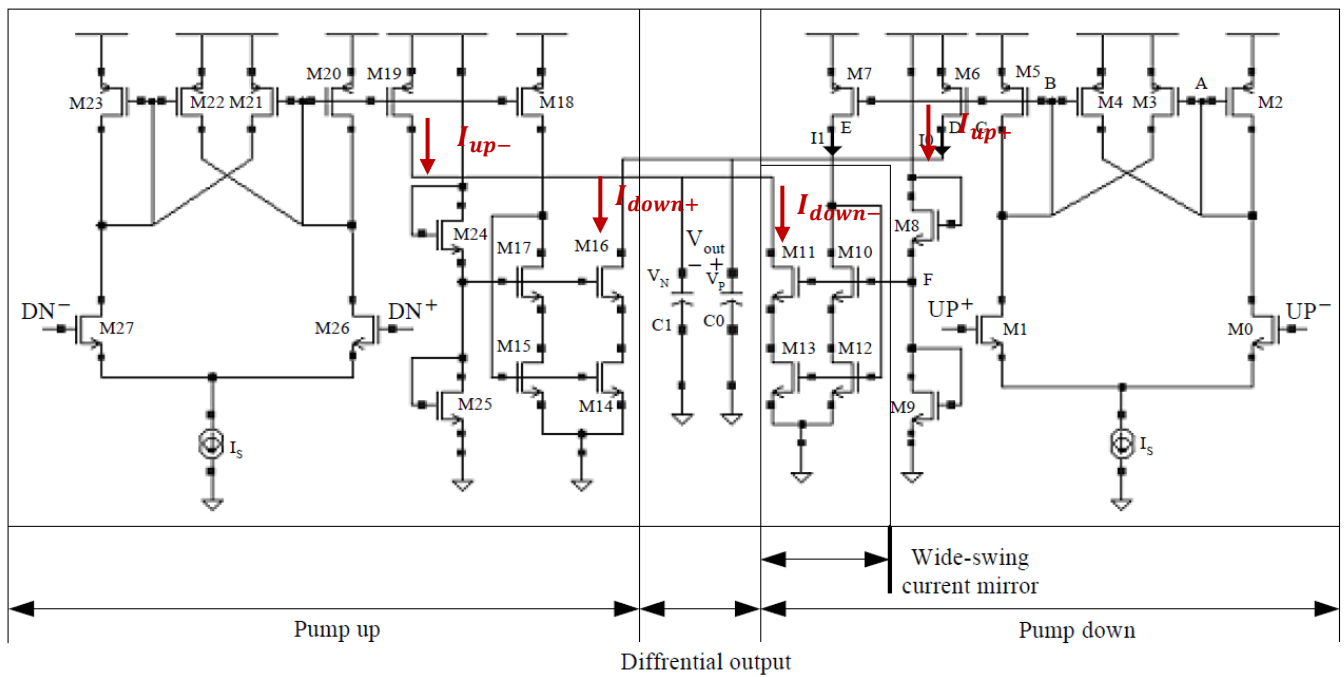
Lab 08

Fully Differential Charge Pump

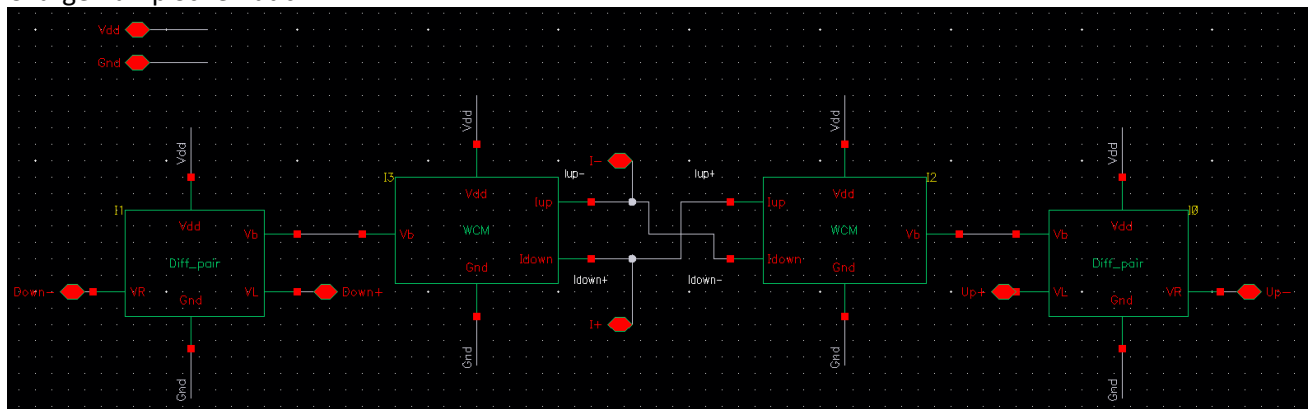
In this lab, we will design a fully differential charge pump



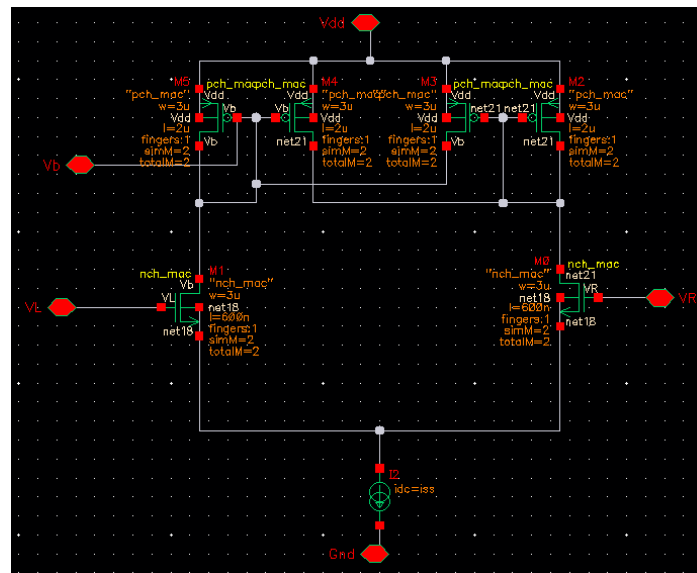
1- Schematic of charge pump



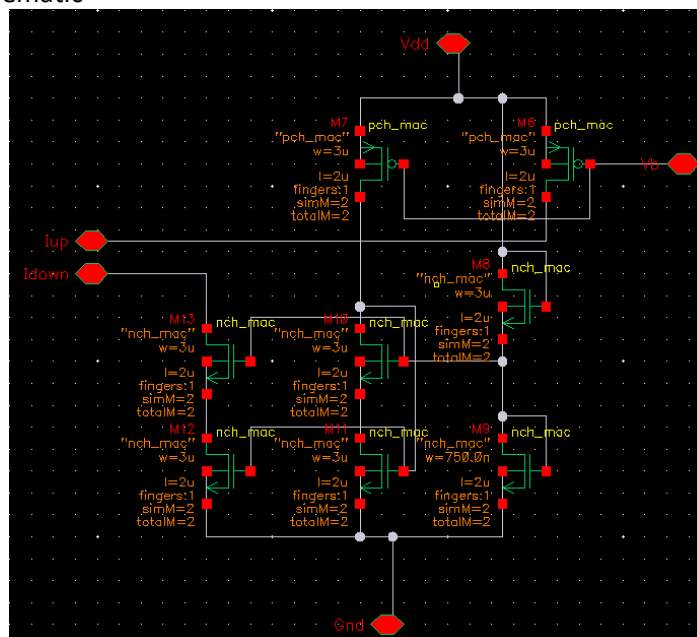
Charge Pump Schematic



Diff. Pair Schematic



Wide Current mirror Schematic



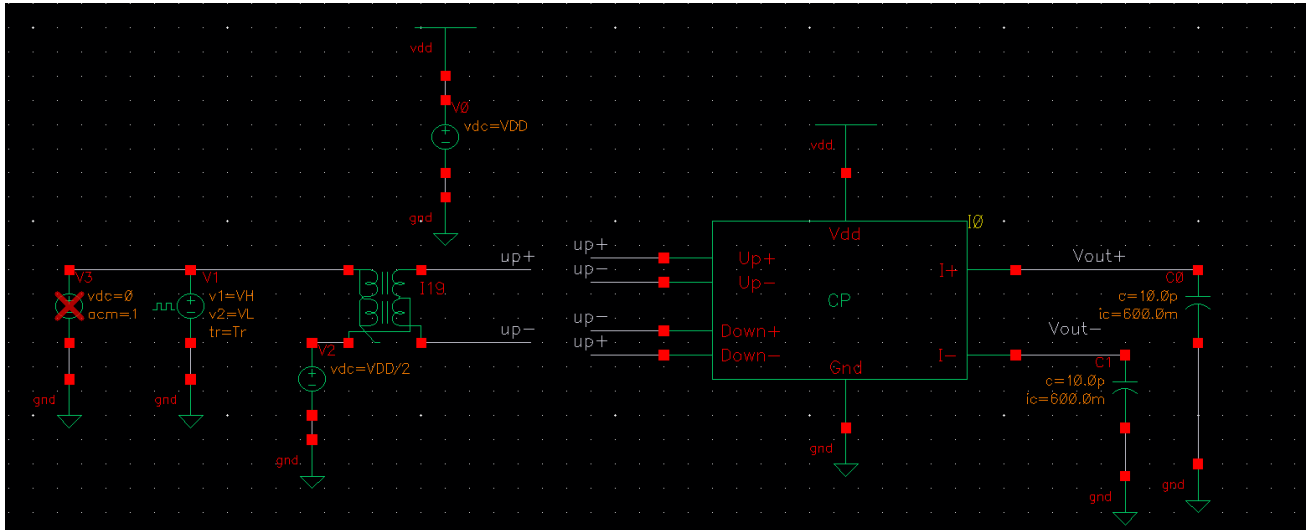
Use these Parameters in Lab simulations:

Iss	40u A
Vdd	1.2 volt
All Tr.s W/L expt the following	3u/2u,M=2
Diff.pair inputs	3u/0.6u,M=2
M9,25	0.75u/2u
C0,C2	10p
Period of Clock	800n
Pulse width	0.5*Period
Rais time / fall time	PW/10
Caps initial values	0.6 Volt

*We will test this charge pump is state 10 and 01 so $UP = \overline{Down}$
So, in differential case, $Up+ = Down -$ & $UP-=Down+$*

2- Transient analysis

We use our input source as the following



Where $V_H = V_{DD} = -V_L$.

Run transient analysis to $t=10u$

in tran window chose options set outstart time to be 4.8u (to skip start up)

Required plots, Vout , I+, I-, Iup (Iup+ - Iup-), Idown (Idown+ - Idown-)

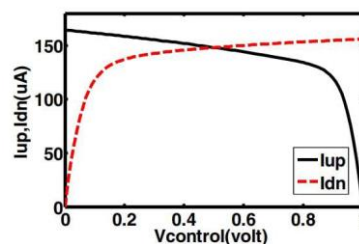
*Note: when you measure I_{up} it will be negative to (current flow out of the node)
also I_{up} -

so $I_{up} = I_{up-} - I_{up+}$

3- Outputs variations

In this part, we want to see the relation bet output Voltage (will be the control voltage to the Vco) and the current in both the Up and down Branches

the following figure is for Single-ended Charge pump



From that figure can you expect the curves of the differential circuit?

Use the function **WavevsWave** to plot I_{up} & I_{down} [on y axis] Vs V_{out} (V_{cont}) [on X axis] and measure the variations in each one of them.

*****in tran window chose options set outstart time to be 4.8u to t=5.2 u (to see cycle[01 state])
and Repeat from 5.2 u to t=5.8 u (to cycle[10 state])***

[Expected 2 figures one for 01 state and one for 10 state]

**in tran window chose options set outstart time to be 4.8u to t=5.6 u (to see cycles[10,01 states])*
[Expected 1 figure for 01,10 state]

**to be able to observe that make 2 sub windows on contain the Transient analysis and another contain WavevsWave outputs*

What you see is due to the similarity of the circuit which it's differential swing has the same behavior but in the inverse sign.

4- Open Loop Gain

Using the AC analysis plot The open Loop Gain in dB and Phase

***note:** don't forget to use a DC Source with AC magnitude = 1.

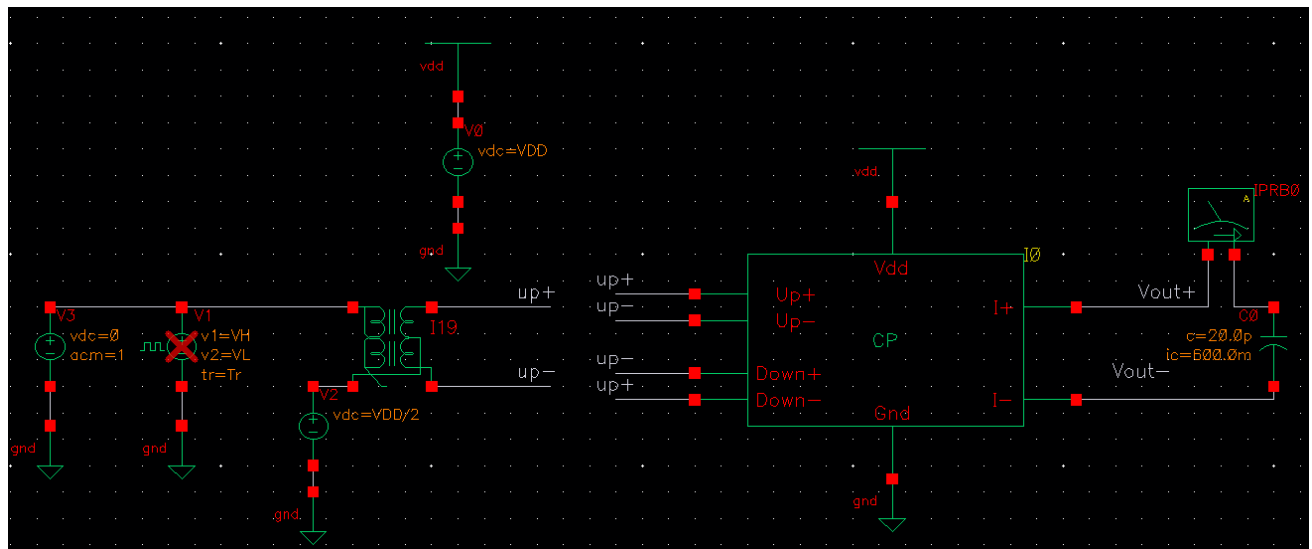
**** Repeat for changes in VDD an Temp**

VDD= 1.08 to 1.32 volt & temp = 0 to 85

- What is the range of variations?

5- Output Current noise

Using the Noise analysis plot Output noise current in dB



***Convert the caps into a Differential Cap**

6- Current mismatch:

As you observed in the part 2 and 3 there is a difference in the values of up and down current.

In this part we will measure the Difference between **lup** and **ldown** then run **monte-carlo**.

***you can measure the max values of lup and down in trans. analysis (with one cycle only as in part 3)**